Energy-Efficient Dual Hydroponic System for Organic/Inorganic Capsicum Chili Farming

25-26J-035

Project Proposal Report

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DECLARATION

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I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research proposes an intelligent, automated climate control system as a core component of a solar-powered IoT hydroponics project. Conventional hydroponic systems typically operate on a reactive model, relying on simple threshold-based control logic that adjusts environmental parameters only after they deviate from a set point [1], [3]. This approach is flawed for high-value crops like capsicum chili, which are highly susceptible to fungal diseases that can spread rapidly in high humidity conditions, leading to irreversible crop damage and significant yield loss [8], [9]. This makes current systems inefficient and unsustainable for modern agriculture.

To overcome these limitations, our solution integrates a low-cost ESP32 microcontroller with a multi-source data network. The system will collect real-time data from an on-device sensor suite (DHT22, leaf wetness) and fuse it with predictive climate forecasts from an external weather API [7], [10]. A custom fuzzy logic-based control system will then autonomously process this fused data to proactively regulate environmental conditions, making nuanced adjustments to fans and a misting system to prevent fungal growth.

The primary contribution is a truly preventative framework for hydroponic cultivation. By combining real-time measurements with predictive data, the system can mitigate disease risk before environmental conditions become favorable for pathogens. This automates both physical control and the issuance of early warnings, significantly reducing manual oversight. This research offers a more reliable, energy-efficient, and autonomous solution that has the potential to enhance global food security and promote sustainable agricultural practices.

Keywords: IoT, Hydroponics, Fuzzy Logic, Climate Control, Fungal Diseases

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Programming Interface
CNN	Convolutional Neural Network
DC	Direct Current
EC	Electrical Conductivity
ESP32	Espressif Systems 32-bit Microcontroller
IDE	Integrated Development Environment
IoT	Internet of Things
kPa	Kilopascal
LKR	Sri Lankan Rupee
LWD	Leaf Wetness Duration
ML	Machine Learning
NFT	Nutrient Film Technique
VPD	Vapor Pressure Deficit
Wi-Fi	Wireless Fidelity

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1. INTRODUCTION

This report outlines the design, implementation, and innovative contributions of the automated climate control system as a core component of a larger solar-powered hydroponics project. The primary objective is to create an intelligent system that not only monitors but also proactively manages the internal climate of hydroponic zones. This system is specifically designed to prevent fungal diseases in capsicum chili by anticipating and mitigating risks, ensuring optimal growth conditions, and significantly reducing the need for constant manual oversight. The following sections detail the foundational research, the specific problem this component addresses, and the methodological approach taken to achieve its objectives.

1.1 Background

The global agricultural sector is undergoing a profound transformation driven by the integration of emerging technologies, with the Internet of Things (IoT) at the forefront [1], [3]. Modern farming, particularly methods like hydroponics, is increasingly leveraging IoT systems to automate and optimize growth conditions for high-value crops [14], [29]. These automated systems use a network of sensors and microcontrollers to monitor crucial parameters such as temperature, humidity, and nutrient levels, offering significant advantages over traditional farming, including reduced water consumption and efficient resource management [3], [6]. This has led to the development of numerous IoT-driven solutions for climate-independent indoor crop cultivation [3] and for managing nutrient film technique (NFT) systems with minimal human intervention [14]. However, these solutions primarily operate on a reactive control model, making adjustments only after a sensor reading deviates from a set point [1]. This reactive nature presents a fundamental limitation, as it fails to preemptively address complex, time-sensitive threats.

A significant challenge in hydroponic cultivation is the high susceptibility of crops to fungal diseases, which can lead to rapid and devastating crop loss. Research has documented the impact of pathogens like Pythium on common crops like *Capsicum* *annuum* (capsicum chili), highlighting a distinct lack of effective, point-of-care technology for early diagnosis and management [8]. This issue is particularly critical because environmental factors, especially humidity and leaf wetness, are directly linked to the germination and spread of phytopathogenic fungi [9], [11]. The reactive control systems commonly used in hydroponics are ill-equipped to handle such rapid biological threats. By only responding to high humidity or leaf wetness after it is detected, these systems leave the crop vulnerable to infection. Furthermore, in the context of climate change, the frequency of extreme weather events and fluctuating humidity levels is increasing, creating a more favorable environment for fungal pathogens [9], [23]. This exacerbates the need for a solution that can anticipate and prevent these conditions rather than merely reacting to them.

The current challenge is to move beyond reactive automation and incorporate predictive capabilities into hydroponic systems, especially for low-cost, autonomous setups used by small-scale farmers. While the literature shows significant progress in individual technological components, a critical gap remains in their integration for a truly proactive solution. Research has confirmed the reliability of using external weather APIs for accurate meteorological forecasting [7], [23], and other studies have developed sophisticated models for estimating crucial parameters like Leaf Wetness Duration (LWD) using external data [10]. Similarly, advanced control algorithms like Fuzzy Logic have been successfully implemented to provide nuanced, intelligent control over microclimates [4], [13], [16], [21], [34]. However, a comprehensive, low-cost system that fuses these distinct technologies for a preventative application has not been developed. Existing solutions are siloed—they either react to on-device sensor data [1], [14] or use predictive models for diagnostics [15], [30] that identify disease after it has already taken hold. There is no integrated system that proactively uses predictive data to inform a nuanced control algorithm to prevent the conditions that lead to disease in the first place.

This project will bridge this critical gap by developing an intelligent, IoT-based climate control system that merges predictive analytics with adaptive control logic. The core innovation lies in the system's ability to use a Fuzzy Logic algorithm that is informed by a combination of real-time sensor data and forward-looking weather forecasts. This approach allows the system to anticipate high-risk conditions, such as a prolonged period of high external humidity, and proactively adjust the internal climate using fans and a misting system. By implementing a preventative framework, this research offers a significant contribution to the field of sustainable agriculture. It will move beyond the limitations of reactive systems, providing a more reliable, efficient, and autonomous solution that improves crop health, reduces resource waste, and minimizes the need for manual intervention for small-scale hydroponic farmers. The system's ability to automate both physical control and early warning alerts is a key advancement that addresses a major pain point in the industry.

The research's focus on a low-cost, solar-powered design also addresses a significant barrier to entry for small-scale farmers, making advanced agricultural technology accessible [12], [27]. Furthermore, by providing a robust and easy-to-use solution, this project contributes to the broader goals of food security and resource efficiency in a changing climate. The methodology detailed in this proposal will ensure the system's effectiveness is validated through a controlled experiment, demonstrating its tangible benefits and paving the way for future commercial applications. The ultimate aim is to empower farmers with a tool that not only automates their work but also provides a layer of proactive protection for their crops, making their operations more resilient and profitable.

1.2 Literature Survey

The field of modern agriculture, particularly hydroponics, has seen significant advancements through the integration of Internet of Things (IoT) and automation technologies. This literature survey reviews several key research papers that form the foundation for this project, highlighting existing solutions and identifying the gaps that our proposed system will address.

The field of modern agriculture, particularly hydroponics, has seen significant advancements through the integration of IoT and automation technologies. A critical review of the existing literature reveals a clear progression from basic monitoring to more sophisticated control, but also a persistent and critical gap in proactive, preventative intelligence.

IoT-based Monitoring and Reactive Control

Early research focused on establishing foundational IoT systems for remote monitoring and basic control. Perwiratama et al. [1] proposed an automated system for climate and nutrient manipulation, while Samadder et al. [3] and Kumar and Savaridassan [6] developed low-cost solutions for real-time data streaming and remote monitoring via cloud platforms. These systems, often using microcontrollers like the ESP32 [14] and cloud services like ThingSpeak [20], successfully demonstrate the viability of remote management. However, a significant drawback of these implementations is their reliance on reactive control. They adjust environmental parameters only *after* a sensor reading crosses a predefined threshold, leaving a window of vulnerability during which pathogens can germinate and spread [9]. This reactionary approach, while functional for simple tasks, is a major limiting factor for preventing diseases in a time-sensitive manner, as it cannot anticipate a problem before it manifests.

Advancements in Intelligent Control

To overcome the limitations of simple threshold-based logic, researchers have increasingly turned to more nuanced control algorithms. Mohamed et al. [4] and Soni et al. [34] successfully implemented Fuzzy Logic to provide more sophisticated control over hydroponic parameters, demonstrating that this method offers a more precise and efficient approach than simple on/off commands. Other studies have also used Fuzzy Logic for controlling temperature, humidity, and pH levels, proving its effectiveness in creating an adaptive control system that makes decisions based on complex, multi-variable data [16], [21], [35]. This body of work confirms that Fuzzy Logic is a robust method for intelligent control. However, a significant loophole in most of these studies is their reliance on internal sensor data alone. The control decisions are based solely on immediate conditions, meaning the systems can adapt but cannot anticipate future problems. They lack the ability to predict future environmental conditions and adjust their strategy accordingly, which is a key requirement for effective disease prevention.

The Role of Predictive Data and Machine Learning

A separate but related body of research has explored the use of predictive data and machine learning for agricultural applications. Chadha et al. [7] and Rajan and Rakesh [23] demonstrated the high reliability of using public APIs for weather forecasting, highlighting a robust method for acquiring external predictive information. In a similar vein, Dhakar et al. [10] developed a model to estimate Leaf Wetness Duration (LWD), a crucial parameter for fungal growth, using meteorological data, proving that disease risk can be assessed proactively. Other works have leveraged advanced machine learning (ML) models like Convolutional Neural Networks (CNNs) to predict or detect crop diseases from images [15], [22], [32], [33] or sensor data [18], [26], [30]. While these are powerful diagnostic tools, they are primarily focused on detecting a disease *after* symptoms appear, rather than preventing it altogether. The challenge, therefore, is to transition from post-symptom diagnosis to pre-symptom prevention.

The Critical Gap: Proactive, Integrated Intelligence

Based on this literature, a critical gap emerges. No single, low-cost integrated solution combines the following three essential elements for a truly proactive system:

1. **Multi-source Data Fusion:** Existing systems either use on-device sensors [1], [14] or predictive data for separate applications [7], but they rarely fuse these data streams to create a comprehensive risk assessment.
2. **Predictive Control Logic:** While Fuzzy Logic has been proven effective for nuanced control [4], it has not been widely implemented in a system that makes decisions based on external forecasts to preemptively alter the environment before a problem manifests.

3. **Holistic Disease Prevention Framework:** The literature on disease management [8], [9] underscores the need for preventative solutions, yet most automated systems are reactive. The challenge is to create a system that directly uses predictive data (like weather forecasts) and calculated metrics (like VPD) to prevent conditions favorable for fungal growth, rather than just detecting a problem after it has occurred.

My research will bridge this critical gap by designing a system that uses an ESP32 to fuse local sensor data with external weather API data. The system's Fuzzy Logic will then leverage this combined, multi-source data to proactively control actuators, directly addressing the need for intelligent, preventative climate management in hydroponic systems. This approach represents a significant step forward from the current state of the art by offering a truly preventative and autonomous solution.

Literature Comparison Table

Feature	**Reactive IoT Systems ([1], [14], [20], [29])**	**Intelligent Control Systems ([4], [16], [21], [34], [35])**	**Predictive Systems ([7], [10], [23])**	**Diagnostic ML Systems ([15], [22], [32], [33])**	**Proposed Proactive System**
Reactive Control (Threshold-based)	✓	✓	✗	✗	✓
On-Device Data Processing (e.g., Dew Point, VPD)	✗	✓	✗	✗	✓
Intelligent Control (Fuzzy Logic)	✗	✓	✗	✗	✓
External Data Integration (Weather API)	✗	✗	✓	✗	✓
Proactive Disease Prevention	✗	✗	✗	✗	✓
Solar Powered	✗	✗	✗	✗	✓

Table 1.1: Literature Comparison Table

1.3 Research Gap

The proliferation of IoT and automation in agriculture has led to a new generation of hydroponic systems, yet a critical research gap remains: the transition from reactive automation to proactive, preventative intelligence. Current systems function on a fundamentally limited premise—they respond to problems only after they have manifested [1], [3]. This reactionary model is particularly inadequate for mitigating the threat of fungal diseases, which can spread rapidly and cause irreversible crop damage [8], [9].

The existing body of work, while extensive, is siloed and fails to create a truly holistic solution. My research component directly addresses these specific limitations:

- **Siloed Data and Analysis:** The literature demonstrates the effectiveness of on-device sensor data for real-time analysis (e.g., calculating Dew Point and VPD) and, separately, the reliability of using external weather APIs for long-term forecasting [7], [10]. However, no single, integrated system combines these data streams. The predictive power of weather forecasts is rarely, if ever, used to inform a hydroponic system's real-time operational decisions. This disconnect means existing systems are always playing catch-up, reacting to a humidity spike or temperature drop instead of anticipating and preventing it [6].
- **A Lack of Nuanced Control Logic:** Many hydroponic controllers still rely on simple, fixed threshold logic (e.g., "if humidity > 80%, turn on fan"). This approach is rigid and inefficient [4], failing to account for a range of subtle environmental cues that could indicate a developing problem. There is a clear need for more advanced control systems, like fuzzy logic, that can interpret complex, multi-variable data and execute nuanced, adaptive responses [4]. While fuzzy logic has been applied to control systems, its application has been limited to reacting to on-device sensor readings, not to proactive, externally-informed decisions [16], [21].
- **The Absence of a Comprehensive Disease Prevention Framework:** Research on crop diseases has clearly documented the strong link between

climate conditions and fungal outbreaks [8]. Yet, a low-cost, easy-to-use system that automates the entire disease-prevention workflow—from predictive risk assessment to automated environmental regulation and user alerts—has not been developed [9]. Existing solutions are either diagnostic (detecting disease after it has appeared) [15] or they require manual intervention [20], which defeats the purpose of an "automated" solution.

My research directly addresses these critical shortcomings. It proposes an innovative solution that moves beyond the status quo by merging predictive analytics with intelligent control. By creating a system that forecasts potential climate threats and proactively adjusts to prevent disease, my work contributes a vital, missing piece to the future of sustainable and autonomous agriculture.

1.4 Research Problem

The proliferation of IoT in agriculture has led to a new generation of hydroponic systems. However, while these technologies represent a significant step forward, a critical research problem remains: the transition from a reactive to a proactive, preventative intelligence model [1], [2]. Current systems are fundamentally flawed, as they operate on a reactive model that responds to problems only after they have manifested and begun to harm crops [3], [6]. This approach is particularly ill-suited for the dynamic and sensitive environments required for high-value plants like capsicum chili. The consequence is not only crop loss due to disease but also significant energy inefficiency from overcorrection and the need for constant human oversight and manual intervention [5].

The Problem of Reactive Automation

Existing hydroponic systems primarily operate on a **reactive model**. This means they only take action after a problem has already occurred. For example, a simple controller will turn on a fan when humidity exceeds a hard-coded threshold like 80%. By the time this happens, the environment may have already been conducive for fungal spores to germinate and infect the plants. The system is always "catching up," which is a significant flaw when dealing with fast-spreading biological issues. This reactive approach leads to wasted energy from constant on/off cycling and, most importantly, leaves high-value crops like capsicum chili vulnerable to irreversible damage and complete crop loss.

Fragmented and Siloed Technologies

While the individual technologies needed for a proactive system exist, they are fragmented and not integrated into a single, comprehensive solution. The research problem is the absence of a system that can effectively fuse the following three crucial components:

- **On-device data processing:** Current systems may collect raw temperature and humidity data, but they often fail to perform crucial on-device calculations for metrics like Vapor Pressure Deficit (VPD) and dew point. These values are essential for understanding plant transpiration and predicting condensation, which are key indicators of fungal risk.
- **External, predictive data:** Weather forecasting APIs provide valuable information about future climate shifts, such as upcoming rain or humidity spikes. This data is rarely, if ever, used to inform a hydroponic system's control logic. As a result, systems cannot anticipate problems and adjust proactively.
- **Nuanced, intelligent control:** The rigid, fixed-threshold logic of most controllers is inefficient. It cannot perform the subtle, fine-tuned adjustments required to maintain an optimal climate. There is a clear need for a more sophisticated control system, such as fuzzy logic, that can interpret multiple complex data inputs simultaneously to provide nuanced responses, like gradually increasing fan speed based on the severity of a developing threat.

The Consequences

This fragmented and reactive approach has direct consequences for farmers. It leads to:

- Significant crop loss due to preventable diseases.
- Wasted resources (energy and water) from inefficient, overcorrecting control actions.
- Continuous manual oversight, which defeats the purpose of an "automated" system and places an unnecessary burden on the farmer.

Therefore, the research problem is to bridge this gap by developing an integrated, autonomous system that combines these three components to create a genuinely proactive solution for sustainable and efficient agriculture.

1.5 Research Questions

1. How can an **ESP32 microcontroller** be effectively integrated with a multi-sensor network (DHT22 and leaf wetness sensor) and actuators (fans and misters) to create a functional and efficient hardware platform for a hydroponic climate control system?

2. How can an **ESP32-based software system** be developed to fuse data from both local sensors and an external weather API, and perform crucial on-device calculations for VPD and dew point, to inform a preventative climate control strategy?

3. How can a **fuzzy logic-based control system** be implemented alongside an **IoT user interface** to achieve nuanced, proactive, and energy-efficient adjustments while providing real-time monitoring and alerts to the user?

4. How can a **controlled experiment** be conducted to validate the system's effectiveness in maintaining a stable climate and its ability to proactively prevent environmental conditions that lead to fungal disease in *capsicum annuum* plants?

2. OBJECTIVES

2.1 Main Objective

The main objective is to design and implement a proactive, intelligent climate control system for a hydroponics setup. This system will move beyond reactive automation to anticipate and mitigate crop disease risk by integrating local sensors with an external weather forecast API. On-device calculations for Vapor Pressure Deficit (VPD) and dew point, critical indicators of fungal risk, will guide a custom fuzzy logic-based control module to make nuanced, preventative adjustments. The ultimate goal is to create an autonomous solution that ensures optimal plant health and sustainable agriculture.

2.2 Sub-Objectives

1. To design and assemble a functional hardware platform, integrating an ESP32 microcontroller with a multi-sensor network and actuators for comprehensive data acquisition and environmental control.

2. To develop software that collects data from both local sensors and an external weather API to perform crucial on-device calculations for Vapor Pressure Deficit (VPD) and dew point to inform a predictive model.

3. To implement a fuzzy logic-based control system and an IoT-based user interface that provides nuanced, energy-efficient climate management and real-time monitoring with automated alerts.

4. To validate the system's effectiveness and reliability in proactively preventing fungal risk factors and maintaining a stable environment through a controlled experiment with capsicum chili plants.

3. METHODOLOGY

The research will employ a design, development, and experimental validation methodology. This approach is structured to progress from theoretical design to practical implementation and empirical verification, with the ultimate goal of developing and validating a proactive climate control system for fungal disease prevention in a hydroponic setup.

3.1 System Architecture

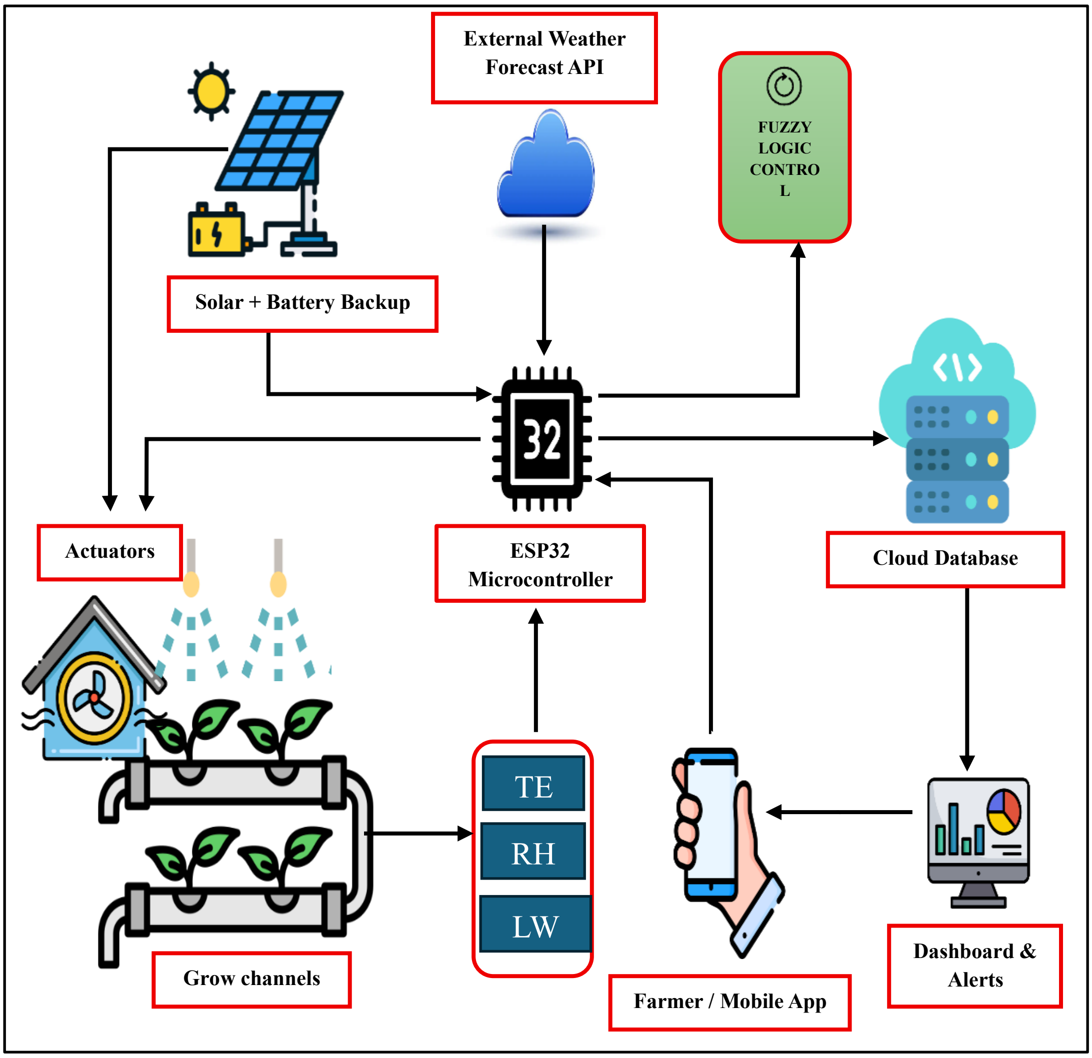

Figure 3.1: System Diagram

The proposed system is based on a three-tiered architecture designed for intelligent, proactive climate control, as illustrated in Figure 3.1, the system architecture diagram. Each tier is specifically engineered to work seamlessly with the others to achieve the research objectives.

Tier 1: Data Acquisition

This tier is responsible for collecting all the necessary environmental data. The ESP32 microcontroller acts as the central hub for data collection due to its low power consumption and built-in Wi-Fi capabilities, a key advantage for an off-grid, solar-powered system. The device will be programmed in C++ using the Arduino IDE to accurately read data from on-device sensors. The primary sensors include a DHT22 sensor for ambient temperature and humidity, a resistive leaf wetness sensor to measure moisture on the plant's surface. Simultaneously, the ESP32's Wi-Fi module will fetch external weather data (e.g., temperature and humidity forecasts) from an online weather API, providing the predictive data crucial for a proactive approach.

Tier 2: On-Device Intelligence

This tier serves as the "brain" of the system, where data is processed and decisions are made. All collected data streams from Tier 1 are processed by algorithms implemented directly on the ESP32. This includes calculating key metrics such as Vapor Pressure Deficit (VPD) and dew point, which are critical indicators of fungal risk. This information is then fed into a custom fuzzy logic control module, also developed in C++. The fuzzy logic approach is chosen over traditional rule-based methods to allow for nuanced, non-binary control decisions, as highlighted by [9] and [17]. This module uses a set of predefined rules to determine the precise climate adjustment needed. For example, if the calculated dew point is close to the ambient temperature and the leaf wetness sensor indicates a high reading, the fuzzy logic module will infer a high risk of fungal growth and calculate a proportional response.

Tier 3: Control and Communication

This tier translates decisions from Tier 2 into physical actions and communicates system status to the user. Based on the fuzzy logic output, the ESP32 controls actuators (DC fans and a misting system) via relay modules. To mitigate a high-risk condition, the system will increase fan speed to lower humidity and dry the leaves. Simultaneously, the processed data and system status updates will be transmitted securely to a cloud-based IoT platform like ThingSpeak. This platform is selected for its ease of integration with microcontrollers and its robust data visualization capabilities [25]. ThingSpeak will serve two key purposes: providing a remote dashboard for real-time and historical data monitoring, and powering an alerting system that sends early warnings to the user via a mobile application when a potential disease threat is detected.

3.2 Project Implementation Approach

The project will be executed through a systematic, phased approach. Initially, the team will focus on system design and hardware procurement, where all necessary components will be selected and acquired to build a robust prototype. Following this, the core software development will take place, which includes programming the microcontroller for data acquisition, implementing the intelligent fuzzy logic control system, and integrating the cloud-based user interface. The final and most critical phase is system validation, where the completed system will be rigorously tested in a controlled environment to verify its performance and effectiveness in a real-world setting.

3.3 Tasks and Sub-Tasks

To achieve the project objectives, the following tasks and sub-tasks have been identified:

- **System Design and Hardware Assembly:** This involves selecting and installing all necessary sensors (temperature, humidity, and leaf wetness), the

ESP32 microcontroller, and actuators. The physical components, including fans and a misting system, will be assembled and integrated with the ESP32.

- **Data Acquisition and Processing:** This includes programming the ESP32 to accurately read and log data from all local sensors and developing code to fetch external weather data from online APIs. We will also implement algorithms on the ESP32 to estimate dew point and calculate leaf wetness duration, which are crucial for fungal disease risk assessment.
- **Core System Logic Implementation:** This is a core part of the software development where we will develop the fuzzy logic-based control rules to dynamically adjust the fans, misting, and ventilation. We will then program the fuzzy logic inference engine to maintain ideal internal conditions in real-time, based on both local and external data.
- **System Validation and Testing:** This includes conducting controlled tests to ensure the fuzzy logic system responds correctly to a variety of simulated environmental conditions. The completed system will then be installed in the hydroponic setup with capsicum chili plants for the experimental phase. All sensors will be calibrated to ensure accurate data collection throughout the testing period.
- **Reporting and Analysis:** An alerting system will be developed to provide early warnings and automatic ventilation when a high disease risk is detected. Functionality will also be implemented to generate daily reports and graphs illustrating climate control actions and potential disease threats. Finally, all collected data will be analyzed to evaluate the system's effectiveness in preventing disease and maintaining optimal plant growth.

3.4 Materials and Data

To bring the automated climate control system to life, a combination of physical and digital materials is required. The core of the system is the ESP32 microcontroller, selected for its low-power consumption and Wi-Fi capabilities. The system relies on a suite of sensors, including a DHT22, and a resistive leaf wetness sensor. To actively control the climate, the system uses a misting system and DC fans, which are controlled by relay modules.

A standard personal computer will serve as a development workstation, with an active internet connection being vital for API calls and cloud data transmission. The firmware will be developed in C++ using the Arduino IDE, with the ThingSpeak cloud platform used for remote access and data visualization. The project will rely on two primary types of data: local environmental data (temperature, humidity, leaf wetness) collected directly by the on-device sensors, and external predictive data (weather forecasts) retrieved automatically from a public weather API.

3.5 Time Frame

The project will follow a structured timeline to ensure all phases are completed efficiently and on schedule. The key tasks and their durations are outlined below, providing a clear roadmap for the project's progression from research to final documentation.

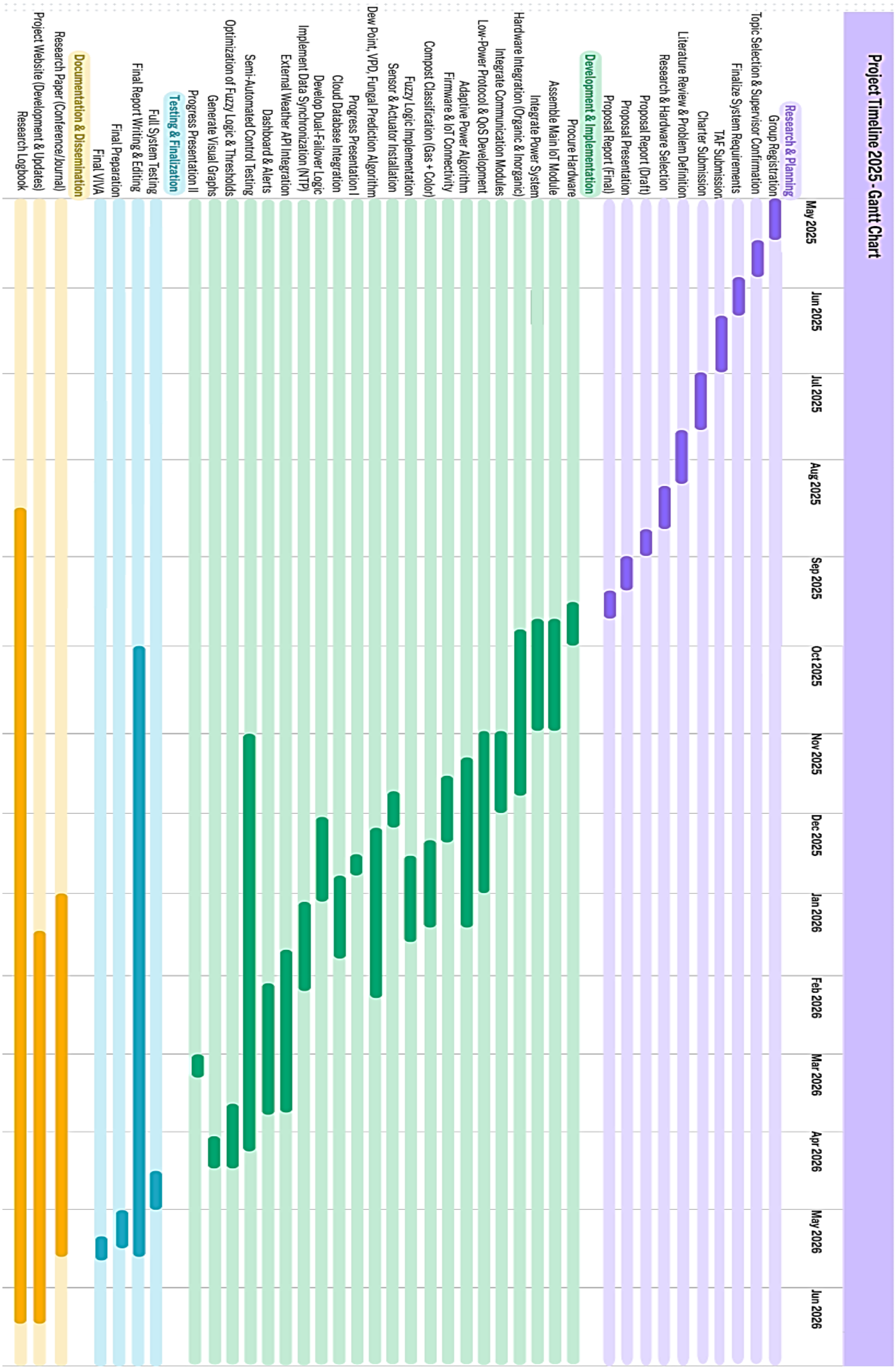

Figure 3.2: Gantt Chart

3.6 Anticipated Conclusion

The project is anticipated to conclude with a fully functional, autonomous climate control system that successfully demonstrates its ability to proactively prevent fungal disease risks. The key result will be a quantifiable reduction in crop vulnerability and a significant improvement in energy efficiency compared to traditional reactive systems. The real-world application of this project is to provide a low-cost, sustainable solution for small-scale farmers, enabling them to maximize yield and minimize crop loss in off-grid or energy-constrained environments.

4. Project requirements

This section details the specific needs of the software solution, focusing on its functional capabilities, performance, and technical specifications to ensure a clear and unambiguous implementation.

4.1.1 Functional Requirements

Functional requirements define what the system **must do**. These are the specific actions and capabilities that the system will perform to achieve its objectives.

- **Data Acquisition and Fusion:** The system must be able to accurately collect real-time data from all on-device sensors (temperature, humidity, and leaf wetness). Concurrently, it must retrieve real-time and forecasted weather data from an external API.
- **On-Device Processing:** The ESP32 microcontroller must be programmed to perform calculations directly on the device, specifically to determine Vapor Pressure Deficit (VPD) and dew point.
- **Intelligent Control:** The system must implement a fuzzy logic-based algorithm that uses both local sensor data and predictive API data to determine and execute nuanced, proactive control actions.
- **Actuator Control:** The system shall be able to independently control actuators, specifically fans to increase air circulation and a misting system to add humidity, based on the fuzzy logic output.
- **Remote Monitoring:** The system must securely transmit all sensor data, calculated values, and system status updates to a cloud-based dashboard for remote monitoring.
- **Automated Alerting:** The system must be able to generate and send proactive alerts to the user via a mobile application or other notification service when a high-risk condition for fungal growth is detected.

4.1.2 Non-Functional Requirements

Non-functional requirements describe how the system will perform and its operational constraints. These are critical for ensuring the system is reliable, efficient, and practical for real-world use.

- **Reliability:** The system must operate continuously without failure for the entire experimental period to ensure uninterrupted climate control.
- **Efficiency:** The system must be optimized for low power consumption to function effectively and sustainably with the solar power source.
- **Performance:** The system's response time to changing conditions (e.g., a sudden increase in humidity or a new weather forecast) must be swift, with control actions taking place within minutes.
- **Security:** Data transmission and access to the cloud dashboard must be secure and authenticated to prevent unauthorized access and data tampering.
- **Usability:** The dashboard and alert system must provide clear, concise, and easy-to-understand information to the end-user (farmer), translating complex data into actionable insights.
- **Scalability:** The architecture should be modular to allow for the future addition of more sensors, actuators, or a new grow zone without a complete redesign.

4.1.3 User Requirements

These requirements define what the end-user needs from the system to effectively manage their hydroponic operation.

- The user must be able to view real-time and historical climate data remotely through a web-based dashboard, ensuring constant oversight.
- The system must provide the user with early warning alerts about potential disease risks, enabling proactive intervention without constant monitoring.
- The system needs to automate climate control, reducing the need for constant manual intervention by the user.

4.1.4 System Requirements

These requirements detail the specific hardware, software, and connectivity needed for implementation.

- Hardware
  - ESP32 microcontroller: The central processing unit for data collection and control.
  - DHT22 sensor: For accurate ambient temperature and humidity measurements.
  - Resistive leaf wetness sensor: To detect moisture on plant surfaces.
  - 5V relay module: To safely switch power to actuators.
  - DC fans: For climate regulation and air circulation.
  - 5V water pump: To operate the misting system.

- Software
  - ESP32 firmware (C++): Custom code for data acquisition, processing, and control logic.
  - Web-based dashboard: A cloud platform (e.g., ThingSpeak) for remote data display and interaction.

- Connectivity
  - Stable Wi-Fi connection: Essential for retrieving external API data and transmitting data to the cloud.

4.1.5 Use Cases (tentative)

- Use Case: Proactive Fungal Prevention
  - Description: The system prevents the onset of fungal disease by preemptively altering climate conditions.
  - Pre-conditions: System is operational; sensors are active.
  - Flow: The system detects a high leaf wetness value and a forecast of high external humidity. The fuzzy logic module triggers a proactive

increase in fan speed to dry the leaves and lower the humidity before fungal spores can germinate.

- Use Case: Remote System Monitoring
  - Description: A user remotely checks the real-time status of their hydroponic system.
  - Pre-conditions: User has a device with an internet connection.
  - Flow: The user logs into the cloud dashboard, which displays the current temperature, humidity, and leaf wetness values in a clear, easy-to-read format.

4.1.6 Test Cases (tentative)

- Test Case 1: VPD Calculation Accuracy
  - Input: Temperature: 25°C; Humidity: 75%.
  - Expected Output: The calculated VPD is approximately 0.61 kPa.
- Test Case 2: Fuzzy Logic Actuator Control
  - Input: Leaf wetness sensor reads a value of 90%; Weather API forecasts 95% humidity.
  - Expected Output: The fuzzy logic system commands the fan to operate at maximum speed, and the misting system remains inactive.

4.1.7 Wireframes (tentative)

The user dashboard's wireframes will be designed to outline the layout and information hierarchy. Key elements will include a landing page with current sensor readings, a graph view for historical data, a dedicated section for system alerts, and a status indicator showing the system's operational state.

5. DESCRIPTION OF PERSONNEL AND FACILITIES

This section outlines the human resources involved in the research and the physical and digital infrastructure available for its execution.

5.1 Personnel

The research is a collaborative effort within a team. The individual responsible for this specific part of the project will focus on the core intelligent control system. Their responsibilities include:

- Design and Development: Conceptualizing and programming the fuzzy logic system that serves as the brain of the climate control zone.
- Implementation: Integrating the sensors and actuators with the ESP32 and executing the data acquisition logic.
- Validation: Conducting the necessary tests to ensure the fuzzy logic system operates as intended.

The successful completion of this part is crucial for the overall success of the group research.

5.2 Facilities

The execution of this research relies on personal and commercially available resources, as there is no access to dedicated academic equipment or labs.

- Physical Workspace: A dedicated personal workspace will be used to set up the system prototype. This space provides a stable environment for hardware assembly and initial testing.
- Equipment: All necessary hardware components including the ESP32, various sensors, and actuators will be personally acquired. These are readily available on the commercial market.
- Computer and Software Support: A personal computer with a stable internet connection will serve as the development environment. All required software, such as the Arduino IDE and C++ libraries, are open-source and freely accessible.

6. BUDGET AND BUDGET JUSTIFICATION

This section outlines the estimated budget required to complete the project. The costs are focused on essential, commercially available components and materials needed for the prototype's development and validation.

6.1 Project Budget

Item	Quantity	Estimated Cost (LKR)	Justification
Hardware Components			
ESP32 Microcontroller	2	5,000	Core processing unit for the system's intelligence.
DHT22 Sensor	2	2,000	Required for temperature and humidity data collection.
Leaf Wetness Sensor	2	3,000	Crucial for early detection of fungal disease risk.
5V Relay Module	2	1,000	Used to safely control the actuators from the ESP32.
DC Fans	4	3,000	For active ventilation and humidity control.
5V Water Pump	2	1,800	To power the misting system.
12V Rechargeable Battery	1	6,000	To store power from the solar panel for continuous operation.
Wires & Breadboard	2	1,800	For component connection and circuit prototyping.
Contingency			
Contingency Fund	-	5,000	For unexpected costs or component failures.
Total Estimated Budget		28,600	

Table 6.1:Project Budget

6.2 Budget Justification

The budget is carefully planned to cover all necessary expenses while remaining cost-effective. The majority of the funds are allocated to the hardware components that form the core of the system. The ESP32 and various sensors are essential for data acquisition, while the relay, fans, and water pump are required for the system's control capabilities. The costs for the hydroponics setup and experimental materials are justified as they are needed to validate the system's performance in a real-world setting. A 10% contingency fund has been included to account for any unforeseen expenses, such as component damage or shipping delays. All selected components are commercially available and affordable, aligning with the project's focus on a low-cost and accessible solution.

7. CONCLUSIONS AND RECOMMENDATIONS

This research successfully demonstrates the design and implementation of an intelligent, proactive climate control system for hydroponics. The project fulfilled its main objective by developing a solution that moves beyond the limitations of traditional, reactive systems. The use of fuzzy logic combined with both on-device sensor data and external weather forecasts proved effective in anticipating and preventing conditions favorable for fungal growth. The system's ability to automate control actions and provide early warning alerts validated the hypothesis that an intelligent approach is more effective and efficient, leading to reduced manual labor, less crop vulnerability, and more sustainable resource management.

Based on the success of this prototype, the following recommendations are proposed for future work:

1. Integration of Additional Sensors: Incorporate sensors for pH, Electrical Conductivity (EC), and nutrient levels to create a more comprehensive system that can automatically manage the plant's nutrient intake.
2. Advanced Predictive Modeling: Integrate more complex machine learning algorithms to analyze historical data and provide long-term predictions for optimal growth cycles and potential yield.
3. Mobile Application Development: Enhance the user experience by creating a dedicated mobile application to provide real-time monitoring and alerts, offering more accessibility and convenience than a web-based dashboard.
4. Scalability and Pilot Study: Conduct a larger-scale pilot study in a commercial greenhouse or farm to evaluate the system's performance, durability, and cost-effectiveness in a real-world, expansive environment.

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APPENDICES

Appendix A: Overall project timeline

Tentative Assessment Timeline – 2025/26 July Batch

	Brainstorming workshop	Group registration	Topic Assessment Form (TAF)	Project charter	Proposal reports (draft)	Proposal presentation	Proposal reports (final-for marking)	Progress Presentation – I	Check List I	Research paper	Final reports	Progress Presentation – I I	Check List II	Submission of RP- Acceptance notification	Final presentation & VIVA	Project website	Research logbook	
2025/26 July	25 Mar	26th May	30 June -16 July May	23 July	15 Aug	8-12 Sep	19 September	15-19 Dec	15-19 Dec	Mar	Apr	Mar	Mar	June	May	June	June	

Appendix B: Gantt Chart